

MOLCHANOVO, Russian Federation

WMO: 293320

Lat: 57.576N

Lon: 83.760E

Elev: 369

StdP: 14.50

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.8	(c) -27.8	(d) -38.5	(e) 0.6	(f) -32.6	(g) -33.6	(h) 0.8	(i) -27.7	(j) 19.0	(k) 14.5	(l) 17.8	(m) 11.3	(n) 2.8	(o) 290	(p) 0.591

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	83.6	65.8	80.8	64.7	77.8	63.6	68.9	79.0	67.3	77.0	65.5	74.2	5.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.1	94.2	74.0	63.6	89.1	71.6	62.1	84.5	70.1	33.4	79.1	32.0	77.4	30.6	73.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 16.7	(b) 14.3	(c) 12.6	(e) -37.6	(f) 87.6	(g) 6.4	(h) 2.5	(i) -42.3	(j) 89.4	(k) -46.1	(l) 90.9	(m) -49.7	(n) 92.2	(o) -54.4	(p) 94.0		
(4) 16.7	(5) 14.3	(6) 12.6	(7) -37.8	(8) 72.2	(9) 6.4	(10) 1.6	(11) -42.5	(12) 73.3	(13) -46.2	(14) 74.3	(15) -49.9	(16) 75.2	(17) -54.5	(18) 76.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.2	-0.2	6.1	23.6	38.0	48.0	63.4	66.3	60.9	48.6	33.8	13.6	6.3		
	DBStd	25.43	14.75	14.07	9.97	8.60	9.77	8.74	5.90	6.25	8.55	9.46	14.58	16.36		
	HDD50	7237	1557	1229	819	370	159	15	0	4	129	509	1093	1353		
	HDD65	11466	2022	1649	1284	810	534	127	58	157	495	969	1543	1818		
	CDD50	1462	0	0	0	11	96	416	505	341	87	6	0	0		
	CDD65	216	0	0	0	0	7	78	98	30	3	0	0	0		
	CDH74	1620	0	0	0	3	74	665	683	178	17	0	0	0		
	CDH80	323	0	0	0	0	12	153	137	21	0	0	0	0		
	WSAvg	6.1	6.4	6.3	7.5	7.1	6.5	4.8	4.2	4.5	5.4	6.6	6.8	6.6		
	PrecAvg	18.7	0.9	0.6	0.6	1.0	1.9	2.3	2.4	2.6	1.9	1.7	1.5	1.2		
	PrecMax	24.2	2.0	1.8	1.6	3.3	4.0	4.6	6.1	5.7	3.9	3.5	2.4	4.6		
(15) Precipitation	PrecMin	12.9	0.2	0.0	0.0	0.0	0.6	0.8	1.1	0.3	0.4	0.6	0.4	0.3		
	PrecStd	2.4	0.4	0.4	0.4	0.6	0.8	1.0	1.1	1.4	0.9	0.7	0.5	0.7		
	0.4%	DB	31.0	34.8	45.2	66.2	81.7	87.8	87.4	82.9	76.5	62.1	42.9	33.4		
	2%	DB	26.4	32.2	41.3	60.3	75.2	84.2	83.6	78.9	71.0	56.5	35.9	31.5		
	5%	DB	23.3	29.4	38.8	55.5	70.1	81.2	80.8	75.4	66.2	50.9	33.7	29.1		
(23) Mean Coincident Wet Bulb Temperatures	10%	DB	20.4	25.2	36.4	51.2	64.2	77.7	77.9	72.2	61.8	46.7	31.6	25.9		
	0.4%	MCWB	19.3	23.9	32.2	41.8	51.6	62.9	65.1	61.6	54.0	41.6	30.5	24.6		
	2%	MCWB	25.5	30.3	33.3	47.3	57.3	64.8	67.4	64.7	58.9	47.1	33.6	30.3		
	5%	MCWB	22.2	27.5	33.2	44.4	54.5	63.7	65.8	63.1	55.7	44.4	32.3	28.0		
	10%	MCWB	19.3	23.9	32.2	41.8	51.6	62.9	65.1	61.6	54.0	41.6	30.5	24.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.9	33.0	38.0	51.9	60.8	70.3	72.4	69.3	62.9	50.5	41.5	32.2		
	2%	WB	25.7	30.6	35.7	48.9	57.5	67.9	69.8	66.8	59.5	47.7	34.2	30.5		
	5%	WB	22.1	27.5	33.8	45.4	55.1	66.1	68.1	65.0	57.0	45.0	32.4	28.2		
	10%	WB	19.3	23.4	32.4	42.2	52.6	64.1	66.4	63.3	54.4	42.1	30.5	24.8		
	MCDB	20.3	25.0	36.5	50.4	63.2	74.3	75.3	69.8	60.9	46.6	31.7	25.8			
(35) Mean Daily Temperature Range	MDBR	12.6	12.9	13.8	16.0	16.4	17.7	16.6	15.6	14.0	10.1	10.7	10.7			
	5% DB	MCDBR	14.1	10.8	14.7	21.4	22.6	22.2	20.7	20.1	20.6	16.6	8.3	12.2		
	5% WB	MCWBR	13.9	9.9	10.0	12.6	10.8	8.3	8.2	9.4	10.3	10.6	8.1	12.1		
	5% WB	MCDBR	14.1	10.5	11.9	20.0	21.7	19.2	18.4	16.8	19.6	16.4	8.3	12.0		
	5% WB	MCWBR	14.0	9.9	9.0	12.6	12.2	8.4	8.8	8.4	10.5	11.2	8.3	12.0		
(40) Clear-Sky Solar Irradiance	taub	0.281	0.314	0.361	0.388	0.398	0.402	0.438	0.404	0.364	0.340	0.303	0.281			
	taud	2.000	1.990	1.941	2.086	2.258	2.326	2.210	2.294	2.358	2.263	2.044	1.942			
	Ebn at Noon	168	218	240	258	265	266	251	252	246	213	157	125			
	Edh at Noon	19	30	42	43	39	37	41	35	28	23	18	15			
(44) All-Sky Solar Radiation	RadAvg	216	526	986	1410	1565	1808	1655	1295	832	418	193	124			
	RadStd	24	41	69	110	152	191	135	112	114	38	34	20			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page